15) Data collection and Initial Exploration.

Program:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
np.random.seed(42)
n = 768
df = pd.DataFrame({
    'Pregnancies': np.random.poisson(3, n),
    'Glucose': np.random.normal(120, 30, n).clip(0, 199).astype(int),
    'BloodPressure': np.random.normal(70, 12, n).clip(0, 122).astype(int),
    'SkinThickness': np.random.normal(20, 16, n).clip(0, 99).astype(int),
    'Insulin': np.random.exponential(80, n).clip(0, 846).astype(int),
    'BMI': np.random.normal(32, 7, n).clip(0, 67),
    'DiabetesPedigreeFunction': np.random.gamma(2, 0.25, n),
    'Age': np.random.gamma(5, 6, n).clip(21, 81).astype(int),
    'Outcome': np.random.choice([0, 1], n)
})
print(df.head())
print(df.info())
print(df.describe())
df.hist(figsize=(20, 15))
plt.show()
sns.pairplot(df)
plt.show()
```

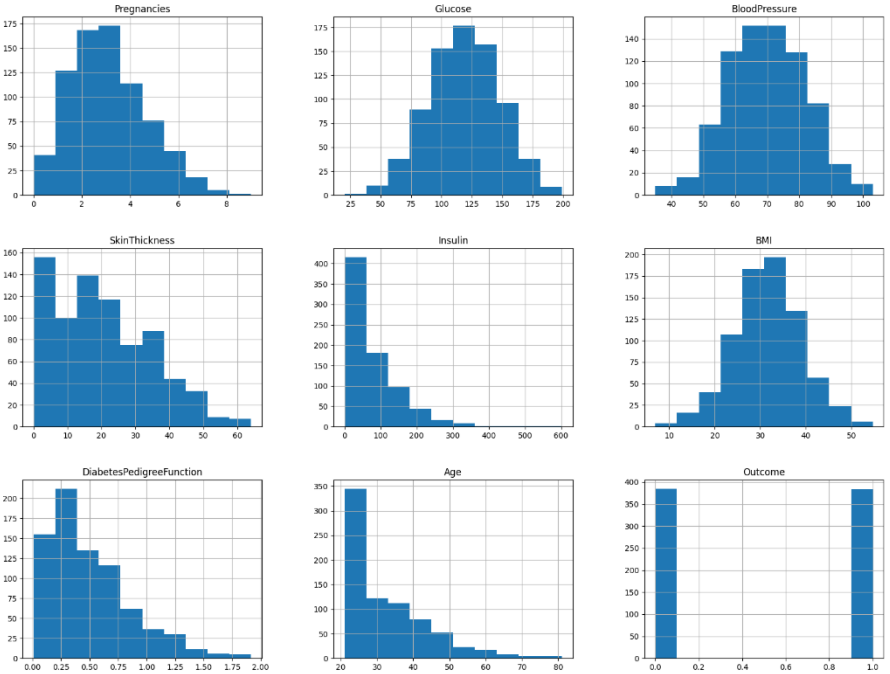
Output:

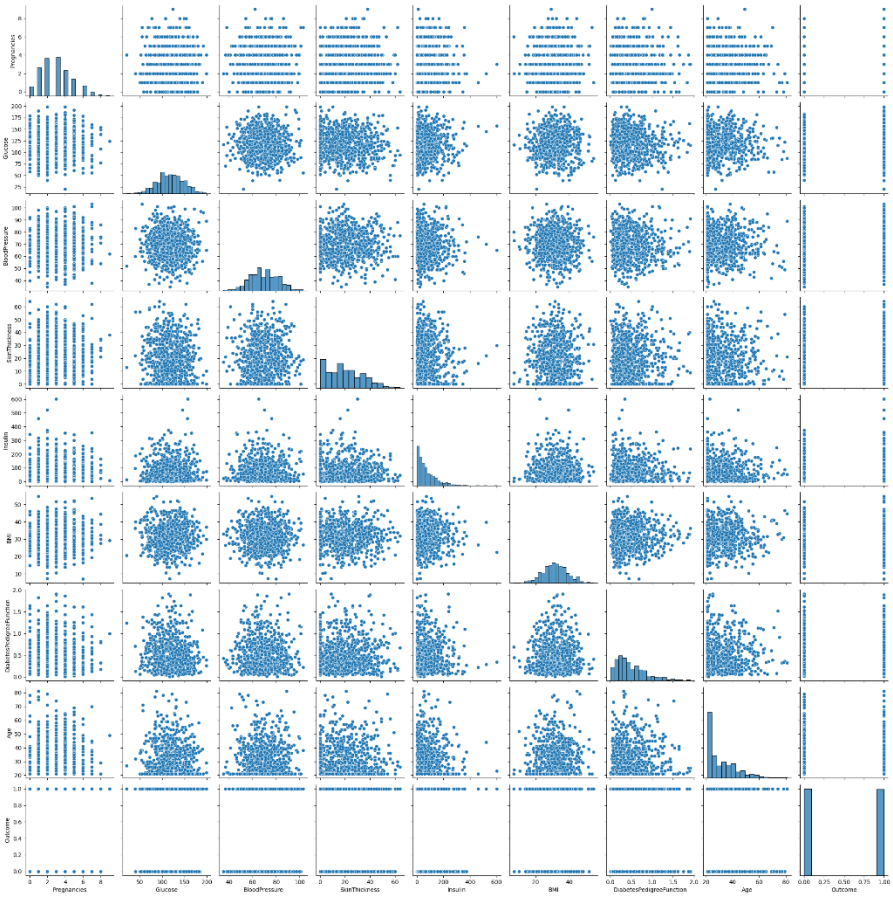
```
   Pregnancies  Glucose  BloodPressure  SkinThickness  Insulin        BMI  \
0            4      165             46             18      353  27.191445
1            1      105             61             46       39  41.748584
2            3      111             78              5      105  29.304742
3            3      104             41              0       92  18.923987
4            2      104             73             20       19  46.095892

   DiabetesPedigreeFunction  Age  Outcome
0                  0.714010   32        0
1                  0.744134   23        0
2                  0.315932   36        1
3                  0.614097   40        0
4                  0.201212   21        0
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 768 entries, 0 to 767
Data columns (total 9 columns):
 #   Column                    Non-Null Count  Dtype
---  ------                    --------------  -----
 0   Pregnancies               768 non-null    int64
 1   Glucose                   768 non-null    int64
 2   BloodPressure             768 non-null    int64
 3   SkinThickness             768 non-null    int64
 4   Insulin                   768 non-null    int64
 5   BMI                       768 non-null    float64
 6   DiabetesPedigreeFunction  768 non-null    float64
 7   Age                       768 non-null    int64
 8   Outcome                   768 non-null    int64
dtypes: float64(2), int64(7)
memory usage: 54.1 KB
None
```

```
       Pregnancies     Glucose  BloodPressure  SkinThickness     Insulin
count   768.000000  768.000000     768.000000     768.000000  768.000000
mean      2.946615  118.786458      69.653646      20.260417   77.225260
std       1.737241   28.937711      11.942598      14.525162   76.767806
min       0.000000   20.000000      35.000000       0.000000    0.000000
25%       2.000000   99.000000      61.000000       9.000000   22.000000
50%       3.000000  118.000000      70.000000      19.000000   55.000000
75%       4.000000  139.000000      78.000000      30.250000  112.000000
max       9.000000  199.000000     103.000000      64.000000  602.000000

              BMI  DiabetesPedigreeFunction         Age     Outcome
count  768.000000                768.000000  768.000000  768.000000
mean    31.591242                  0.499577   31.697917    0.498698
std      7.400237                  0.357731   11.764747    0.500324
min      6.925562                  0.009226   21.000000    0.000000
25%     26.831742                  0.229507   21.000000    0.000000
50%     31.555204                  0.412649   28.000000    0.000000
75%     36.396015                  0.687550   38.000000    1.000000
max     54.680974                  1.914403   81.000000    1.000000
```


Pregnancies
Glucose
BloodPressure
SkinThickness
Insulin
BMI
DiabetesPedigreeFunction
Age
Outcome



Result:

Thus, data collection and initial exploration were performed successfully.